

K716 — CLOSURE: the edge-projection universal factor is RULED OUT (Grace’s sweep failed; Cal caught the backsolve). Clean negative. The projection theory KEEPS its derived segments as narrative. And θ_{13} ’s DERIVE-not-backsolve path is the standard μ - τ breaking SUM RULE — one parameter correlates θ_{23} , θ_{13} , δ , target-innocent, DUNE-testable, no quasi-degeneracy.

Keeper | 2026-07-17 | Casey: “Grace and Cal want us to derive not backsolve the edge projection; think hard on closing the item.” Closing it: the edge factor doesn’t derive (negative closure), and the forward θ_{13} path is the μ - τ sum rule, not the projection.

THE ITEM CLOSED — NEGATIVE (a clean result, per Casey’s mandate)

- Grace’s universality sweep FAILED: across every mixing observable, the boundary→measured factor is within 3% of 1.0 — ZERO near $\sqrt{(2/3)}$, ZERO near $\sqrt{(3/4)}$. The banked forms already match the data at factor 1, including θ_{13} (0.999, not 0.866). The sweep only “passes” by reverse-engineering a boundary value per observable — the per-observable fishing the test was built to catch. Grace caught and refused it.
- Cal caught the backsolve: $\theta_{13} = 3/35 \times \sqrt{(3/4)} = 2/27$ is 0.2% INEXACT (not an identity), and $3/35$ is the unique nearest bare ratio to $(2/27)/\sqrt{(3/4)}$ — the back-solve signature (take the answer, divide by the factor, snap to nearest form). And it’s CIRCULAR: $\sqrt{(3/4)}$ was introduced to explain θ_{13} , so θ_{13} can’t be evidence for it. And it disagrees with the oscillator’s $3/40$.
- VERDICT: the universal face/edge projection FACTOR is RULED OUT. “Structural becomes precision, projected” is NOT supported. The precision/structural split is just that mass forms are sharper than mixing forms — not a projection shear. θ_{13} reverts to its clean exact form $\sin^2\theta_{13} = 1/45 = 1/(N_c^2 \cdot n_C)$, i.e. $\sin^2\theta_{13}/\sin^2\theta_{12} = 2/27 = \text{rank}/N_c^3$ (Cal §17) — banked FORM at 0.1%, mechanism the OPEN piece.

WHAT SURVIVES (real, keep it)

The projection theory as a NARRATIVE unifying the day’s DERIVED segments: - Refraction index $n = N_c/\text{rank} = 3/2 + \text{angle } 41.8^\circ$ (from the Peirce block / muon, predates mixing). - Fresnel- $\sqrt{}$ (amplitude= $\sqrt{\text{intensity}}$) as the ONE $\sqrt{}$ behind every mixing- $\sqrt{}$ (Elie 4693): the refraction $\sqrt{(N_c/\text{rank})}$ sits literally inside $m_u/m_d = \sqrt{(3/14)}$ AND is the projection index — one physical $\sqrt{}$, not two coincidences. - Segment-2 evanescent $V_{cb} = y_t=1$: down at $\sqrt{(2/3)}$, up refracts by $\sqrt{(3/2)}$, $\sqrt{(2/3)} \cdot \sqrt{(3/2)} = 1 = \text{boundary} \rightarrow \text{up evanescent} \rightarrow V_{cb} = \text{down alone}$. Exact self-consistency (Grace/Cal). - The face/edge factors as pure geometry: $\sqrt{((d-1)/d)}$ descent ladder, $\sqrt{(3/4)} \cdot \sqrt{(2/3)} = \sqrt{(1/2)}$ exact composition. These are real geometric facts — they just don’t produce a MEASURABLE universal shear (the sweep showed the data sits at factor 1).

θ_{13} — THE DERIVE-NOT-BACKSOLVE PATH: the μ - τ breaking SUM RULE (standard framework, target-innocent)

Replace the edge-wrap with the published μ - τ breaking sum rule: - $\cos(2\theta_{23}) \approx \sin(\theta_{13}) \cdot \cos\delta$ — ONE breaking parameter drives θ_{23} ’s deviation from maximal, θ_{13} ’s opening, and δ , ALL CORRELATED. (Grimus-Lavoura, hep-ph/0601118; atmospheric sum rules 1308.4314.) - BST’s normal hierarchy ($m_{\nu 1}=0$) selects the C1 branch ($\cos 2\theta_{23} \approx \sin\theta_{13}$), which does NOT need quasi-degeneracy — definitively resolving the K713 spectrum worry. (The C2 branch, $\cos 2\theta_{23} \approx \Delta m^2_{\text{sol}}/\Delta m^2_{\text{atm}}$, would need inverted/quasi-degenerate — NOT BST.) - BST’s banked forms nearly satisfy it: $|\cos 2\theta_{23}| = 2 \cdot (4/7) - 1 = 1/7 = 1/g$, and $\sin\theta_{13} = \sqrt{(1/45)} = 1/(N_c \sqrt{n_C}) = 0.149$; $1/g = 0.143 \approx 0.149$ at $\sim 4\%$ (structural). So θ_{23} (4/7) and θ_{13} (1/45) are consistent with the ONE-parameter μ - τ breaking at structural tier.

Why this is a DERIVATION and not a backsolve (satisfies Cal): 1. It uses ONE breaking parameter (the residual charge-feeling / imperfect $S^4 \mathbb{Z}_2$ restoration), derived from the chargeless-locus geometry — NOT a per-observable factor. 2. The EVIDENCE is the CORRELATION ($\cos 2\theta_{23} \leftrightarrow \sin\theta_{13} \leftrightarrow \delta$), NOT θ_{13} ’s own value

— so θ_{13} is not evidence for its own mechanism (Cal’s circularity rule satisfied). 3. It PREDICTS: the upper octant ($\theta_{23} > 45^\circ$, banked 4/7 [7]) and a specific δ region — a DUNE-testable falsifiable output, the target-innocent evidence Cal’s condition 4 demanded.

THE CLOSURE PLAN ($\theta_{13} + \theta_{23}$ close together, forward)

Derive the ONE μ - τ breaking parameter ε from the chargeless-locus geometry (the residual by which the neutrino isn’t perfectly on S^4). Then the standard sum rule forces $\cos 2\theta_{23} \approx \sin \theta_{13} \cdot \cos \delta \rightarrow \theta_{23}$ (4/7) and θ_{13} (1/45) fall out together at structural tier, correlated, with δ predicted. No projection, no oscillator overshoot, no backsolve. θ_{13} banks at STRUCTURAL tier (not 0.1% precision) with the octant+ δ correlation as the falsifiable evidence. The clean form $1/45 = 1/(N_c^2 \cdot n_C)$ is the target; the sum rule is the mechanism.

HONEST TIERS

- Edge-projection universal factor: RULED OUT (closed negative). Do not resurrect.
- Projection theory segments (refraction, Fresnel- $\sqrt{}$, evanescent V_{cb} , the geometric descent factors): DERIVED, kept as narrative.
- θ_{13} : clean form $1/45 = \text{rank}/N_c^3$ (0.1%); forward mechanism = the μ - τ sum rule (structural, target-innocent, DUNE-testable) — the honest closure path, replacing the backsolve. Banks when the ONE breaking parameter derives from the chargeless locus.
- δ_{PMNS} : now has a home — the same sum rule predicts it (correlated with the octant). A genuine forward prediction.

— Keeper K716, 2026-07-17. Edge-projection universal factor RULED OUT (sweep failed 6/7 at factor 1; edge-wrap a backsolve, Cal). Projection theory keeps derived segments as narrative. θ_{13} reverts to clean $1/45 = \text{rank}/N_c^3$; forward mechanism = standard μ - τ breaking SUM RULE $\cos 2\theta_{23} \approx \sin \theta_{13} \cdot \cos \delta$ (one parameter, correlated, C1/normal-hierarchy, no quasi-degeneracy, DUNE-testable δ). Derive the breaking parameter from the chargeless locus $\rightarrow \theta_{23} + \theta_{13}$ close together forward. See [[Keeper_K715...]], [[BST_Projection_Theory_substrate_to_continuum_2026-07-16]], [[Keeper_K713...]].